MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb-filtered/naveja_recap

generated 2026-08-31 12:30

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	3395
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	4
rdkit_version	2026.03.2
created_at	2026-08-31T09:36:55

B · Composition

quantity	value
molecules	50,933
from coconut	30,121 (59.1%)
from flavordb	20,812 (40.9%)
decompositions	136,414
per molecule	mean 2.68, median 3, max 4
R-group slots	155,642
per decomposition	mean 1.14
k >= 2	12.20% of decompositions

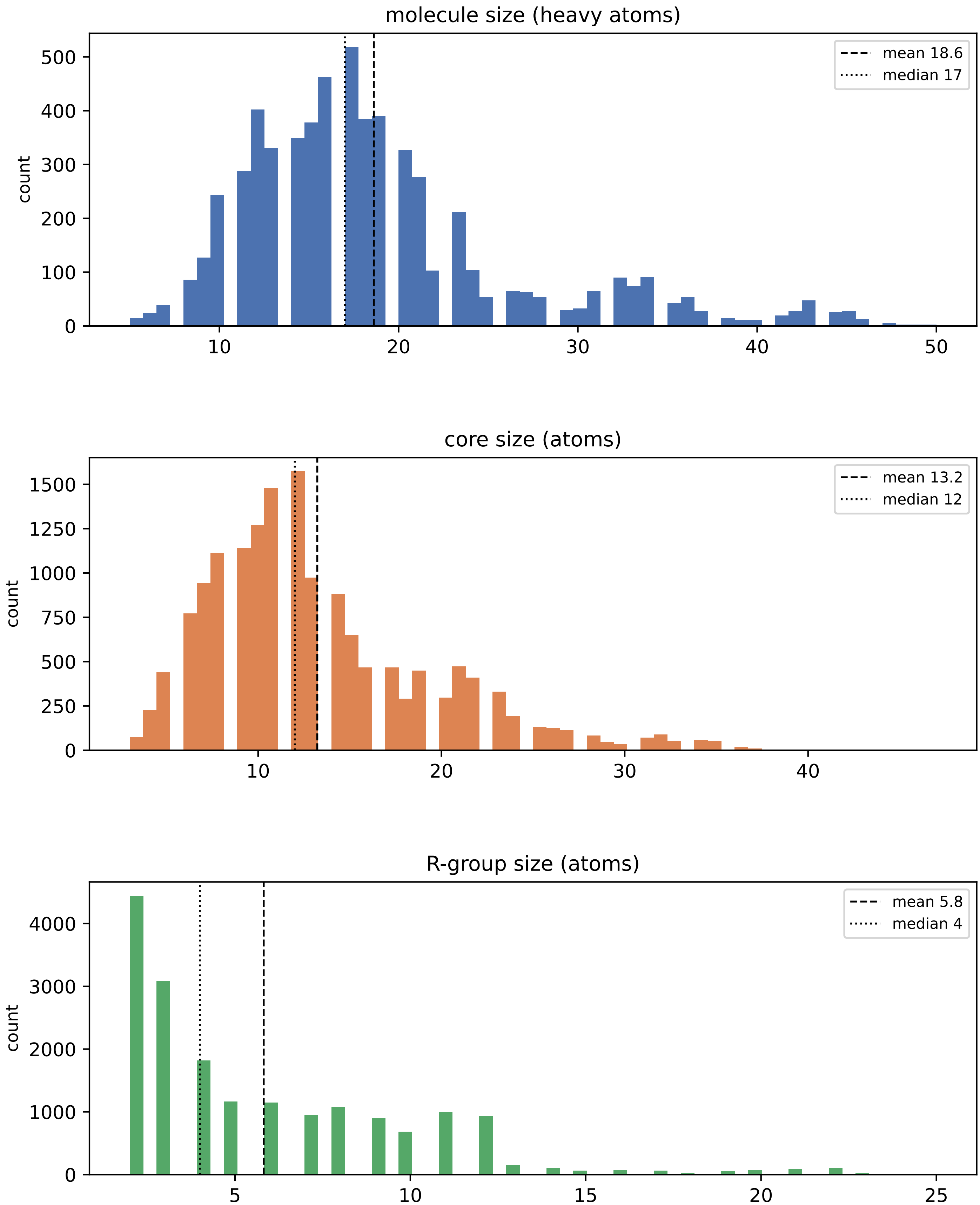
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	50,933 instances/epoch
sampling_unit = decomposition	136,414 instances/epoch (2.7x)
full (mol, core, islinked) space	184,624 (3.6 per molecule)

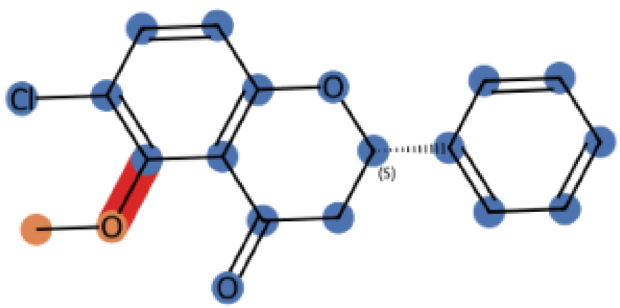
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

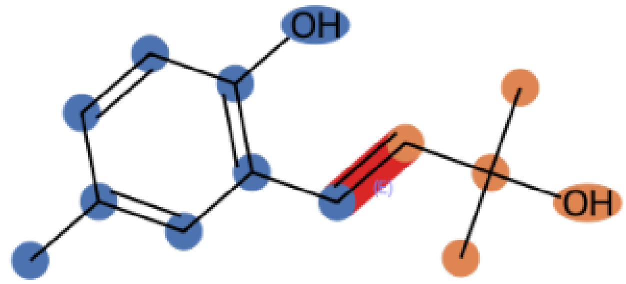


E · Example decompositions (ratio 0.3333333333333333)

CNP0000021   k=1   core 18 atoms, R-groups [2]

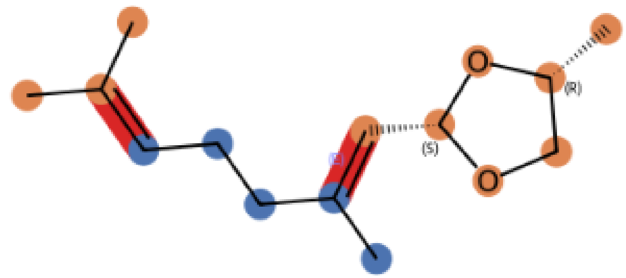


CNP0000061   k=1   core 9 atoms, R-groups [5]

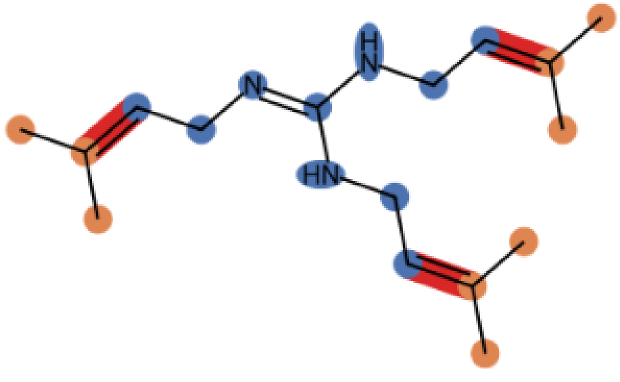


CNP0000308   k=2   core 11 atoms, R-groups [7, 2]

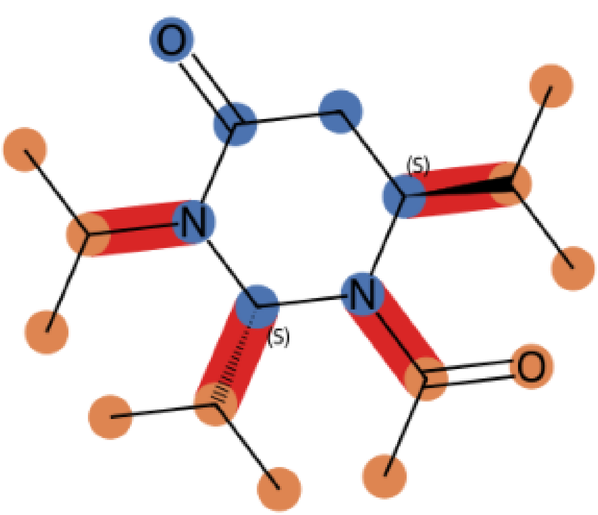
CNP0002830   k=2   core 5 atoms, R-groups [3, 7]



CNP0111421   k=3   core 10 atoms, R-groups [3, 3, 3]

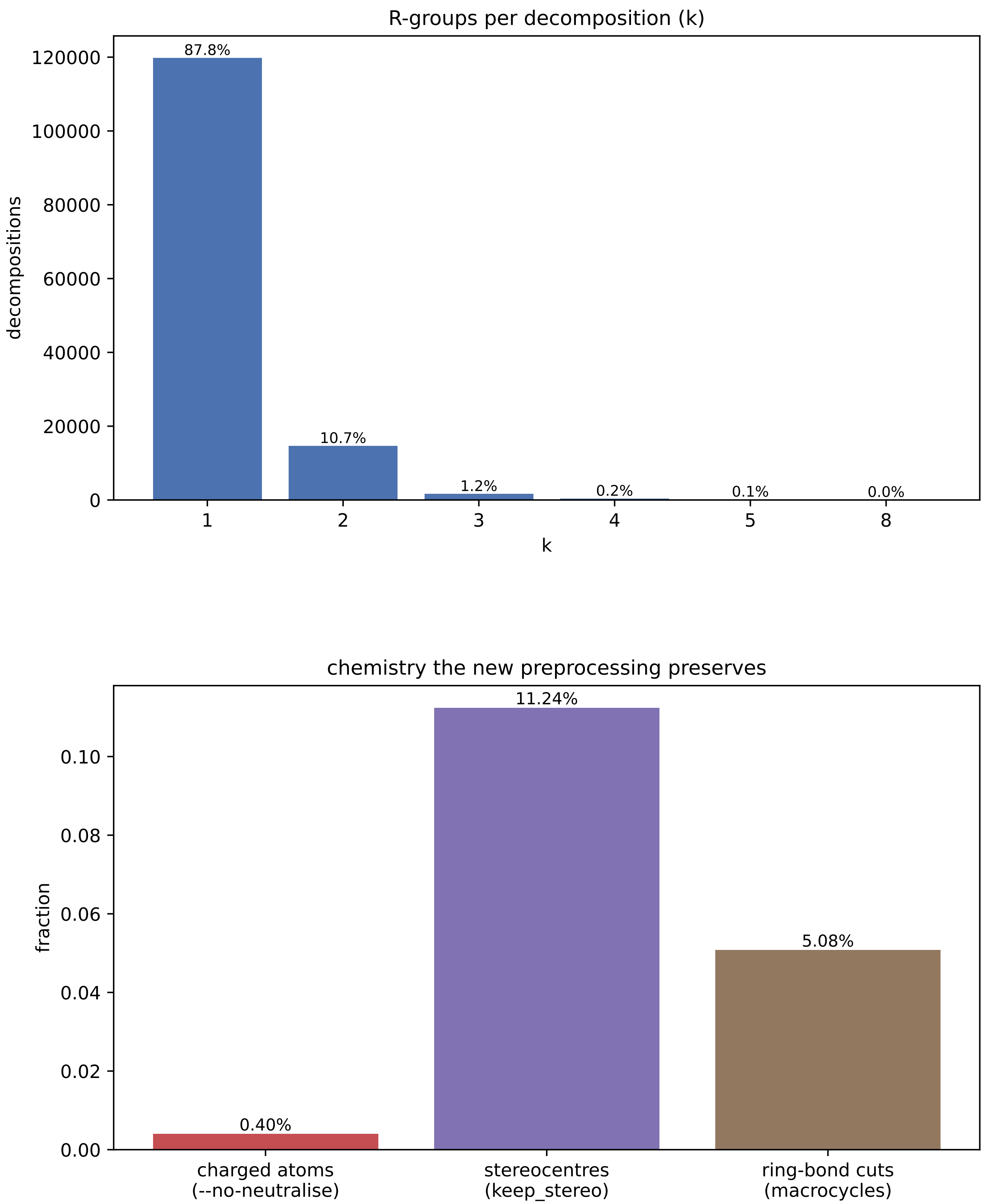


CNP0185661   k=4   core 7 atoms, R-groups [3, 3, 3, 3]

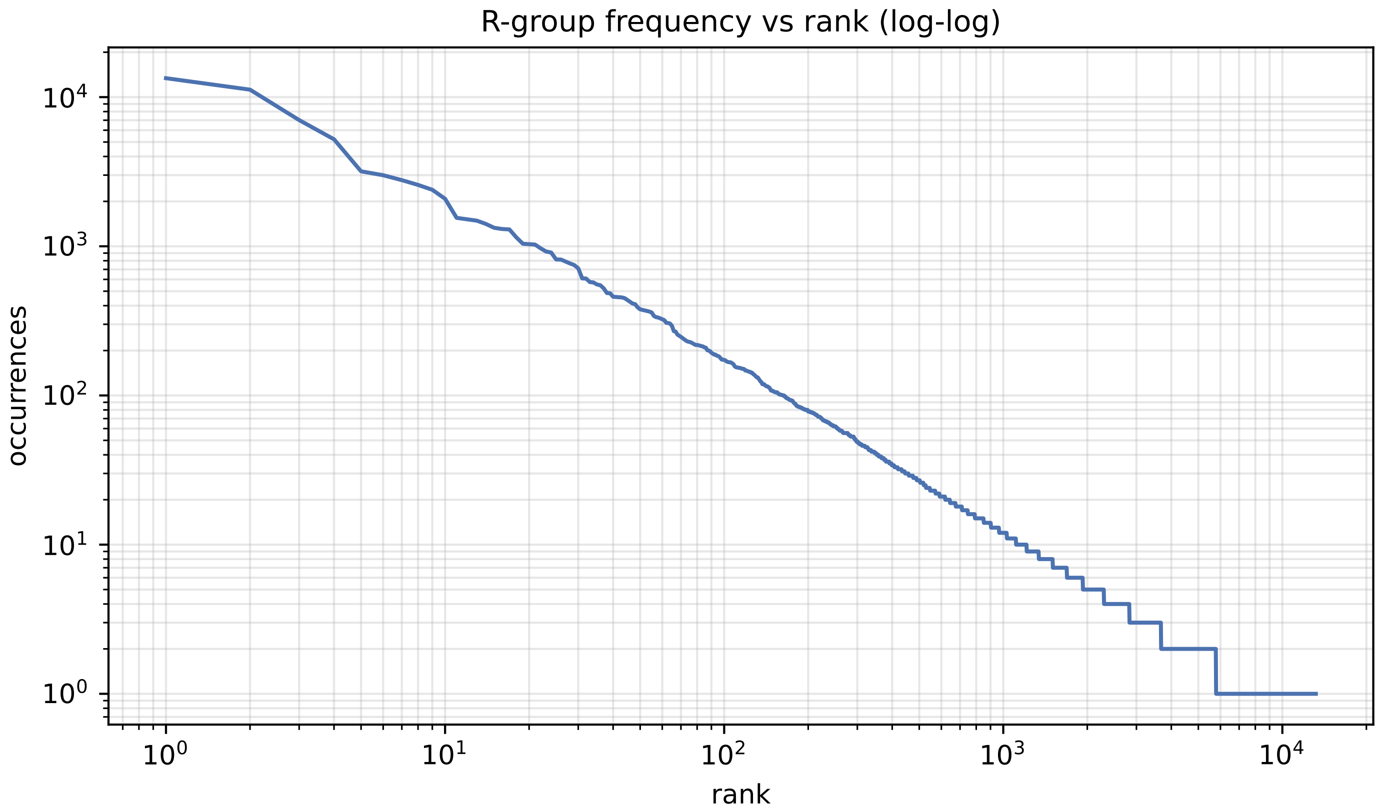


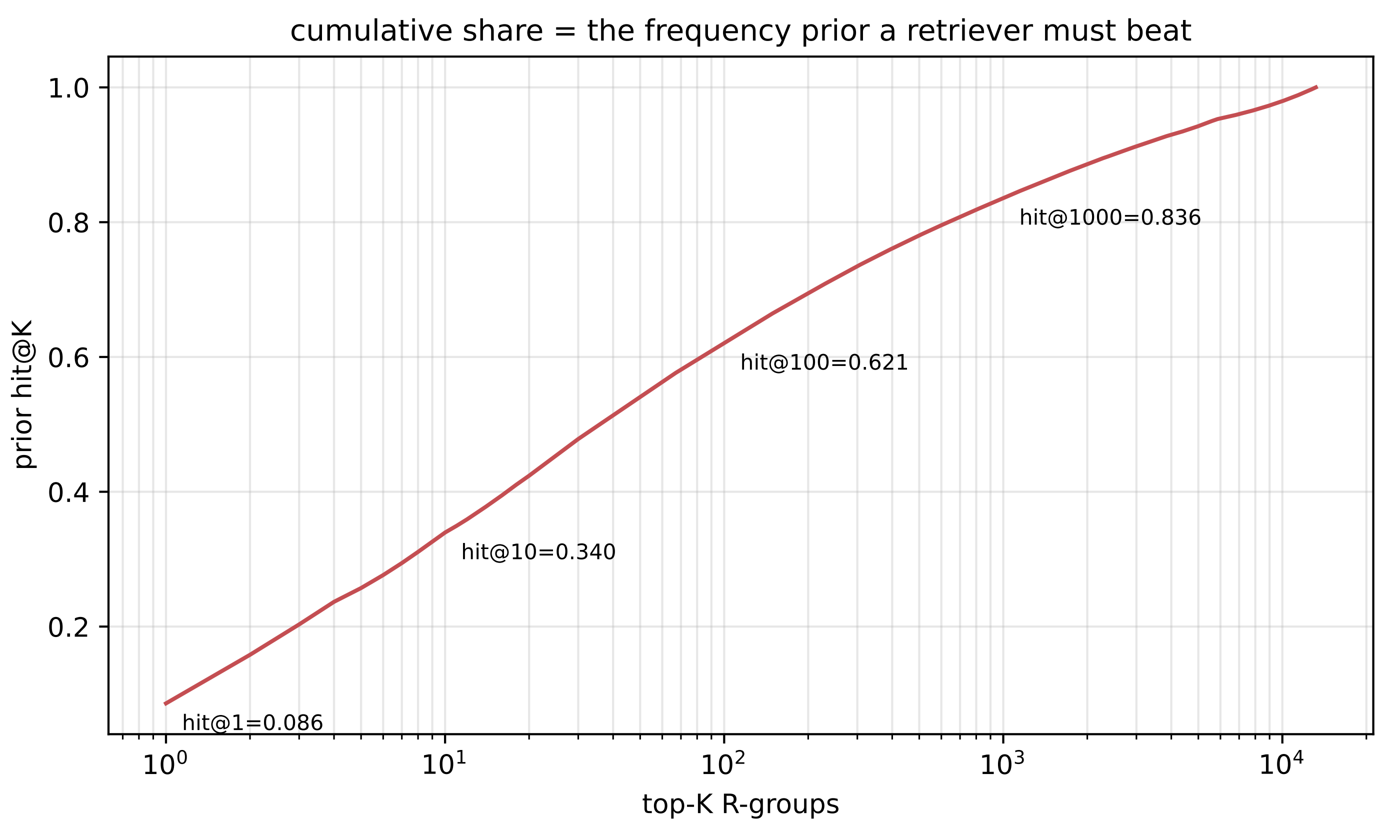
blue = core   orange = R-groups   red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	13,178
occurrences	155,642
effective size (exp H)	467
top-1 share	8.61%
top-10 cover	34.0%
top-100 cover	62.1%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
13,393	8.61%	*OC
11,220	7.21%	*[CH2]O
7,016	4.51%	*[CH2]C
5,215	3.35%	*C(C)=O
3,181	2.04%	*c1ccccc1
2,995	1.92%	*[CH](C)C
2,776	1.78%	*C(C)C
2,579	1.66%	*[CH2]CC
2,392	1.54%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C
2,082	1.34%	*[CH]=O
1,551	1.00%	*[CH]=C
1,514	0.97%	*C(=O)O
1,485	0.95%	*OCC
1,412	0.91%	*[NH]C(Cc1ccccc1)C(=O)O
1,330	0.85%	*C1O[C@H](CO)[C@@H](O)[C@H](O)[C@H]1O
1,306	0.84%	*OC(C)=O
1,297	0.83%	*c1ccc(O)c(O)c1
1,148	0.74%	*[CH2]CCC
1,040	0.67%	*P(=O)(O)O
1,033	0.66%	*O[C@@H]1O[C@@H](C)[C@H](O)[C@@H](O)[C

J · Instance-level common-R-group filter

quantity	value
percentile	99.99
count threshold	9,884
hashes flagged common	2 of 13,178
their share of occurrences	15.8%
single-R-group instances rejected	2,722/18,015 (15.1%)
mean R-group size, rejected	2.00 atoms
mean R-group size, kept	6.50 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.